VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 8: ES6 Basics

Branch/ Class: B.Tech/M.Tech **Date:**
Faculty Name: Prof. S.Gopikrishnan **School:** SCOPE
Student name: **Reg. no.:**

Exercise 1

Write a JavaScript program using ES6 features to calculate the total and average marks of a student. The program should declare variables using let and const, use an arrow function to compute the average, and display the result using template literals. The program should print the student name, total marks, and average marks in a formatted message.

Sample Code Structure

```
let studentName = "Arun";
let mark1 = 85;
let mark2 = 90;
let mark3 = 88;
constcalculateAverage = (m1, m2, m3) => {
   return (m1 + m2 + m3) / 3;
}
let average = calculateAverage(mark1, mark2, mark3);
console.log(`Student Name: ${studentName}`);
console.log(`Average Marks: ${average}`);
```

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JS Course_enroll.js | JS Avg_marks.js

Ex-1 > JS Avg_marks.js > ...

```js
let studentName = "Arun";
let mark1 = 85;
let mark2 = 90;
let mark3 = 88;


let total = mark1 + mark2 + mark3;


const calculateAverage = (m1, m2, m3) => {
    return ((m1 + m2 + m3) / 3).toFixed(2);
};

let average = calculateAverage(mark1, mark2, mark3);

console.log(`Student Name: ${studentName}`);
console.log(`Total Marks: ${total}`);
console.log(`Average Marks: ${average}`);
```

Expected Output
Student Name: Arun
Average Marks: 87.66

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```
PS C:\Users\SYS\Desktop\Web_tecnologies_lab_work\Lab_08> cd Ex-1
PS C:\Users\SYS\Desktop\Web_tecnologies_lab_work\Lab_08\Ex-1> node Avg_marks.js
Student Name: Arun
Total Marks: 263
Average Marks: 87.67
PS C:\Users\SYS\Desktop\Web_tecnologies_lab_work\Lab_08\Ex-1>
```

Exercise 2

Create a JavaScript program that defines an object representing a student with properties such as id, name, department, and marks. Use object destructuring to extract the values and display them. Then create a new object using the spread operator that adds an additional property called grade based on the marks and display the updated object.

Sample Code Structure

```
const student = {
   id: 101,
   name: "Priya",
   department: "CSE",
   marks: 92
};

const {id, name, department, marks} = student;
console.log(id, name, department, marks);
constupdatedStudent = {
   ...student,
   grade: "A"
};
console.log(updatedStudent);
```

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JS Course_enroll.js | JS Stud.js

Ex-2 > JS Stud.js > ...

```javascript
const student = {
    id: 101,
    name: "Priya",
    department: "CSE",
    marks: 92
};

const {id, name, department, marks} = student;

console.log(id, name, department, marks);


const updatedStudent = {
    ...student,
    grade: "A"
};

console.log(updatedStudent);
```

Expected Output
101 Priya CSE 92
{ id: 101, name: 'Priya', department: 'CSE', marks: 92, grade: 'A' }

```
PS C:\Users\SYS\Desktop\Web_tecnologies_lab_work\Lab_08> cd Ex-2
PS C:\Users\SYS\Desktop\Web_tecnologies_lab_work\Lab_08\Ex-2> node Stud.js
101 Priya CSE 92
{ id: 101, name: 'Priya', department: 'CSE', marks: 92, grade: 'A' }
PS C:\Users\SYS\Desktop\Web_tecnologies_lab_work\Lab_08\Ex-2>
```

Exercise 3

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Develop a JavaScript program using ES6 classes and promises to simulate a simple course enrollment system. Create a class named Course with a constructor to initialize the course name and instructor. Include a method to display course details. Implement a promise that checks whether seats are available and prints either "Enrollment Successful" or "Course Full" based on the condition.

Sample Code Structure

```
class Course {
constructor(courseName, instructor) {
this.courseName = courseName;
this.instructor = instructor;
   }
displayCourse() {
console.log(`Course: ${this.courseName}, Instructor: ${this.instructor}`);
   }
}
let course1 = new Course("Web Technologies", "Dr. Kumar");
course1.displayCourse();
let enrollCourse = new Promise((resolve, reject) => {
   let seatsAvailable = true;
   if(seatsAvailable)
resolve("Enrollment Successful");
   else
reject("Course Full");
});
enrollCourse
.then(msg => console.log(msg))
.catch(err => console.log(err));
```

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JS Course_enroll.js

Ex-3 > JS Course_enroll.js > ...

```javascript
class Course {
    constructor(courseName, instructor) {
        this.courseName = courseName;
        this.instructor = instructor;
    }

    displayCourse() {
        console.log(`Course: ${this.courseName}, Instructor: ${this.instructor}`);
    }
}


let course1 = new Course("Web Technologies", "Dr. Kumar");


course1.displayCourse();


let enrollCourse = new Promise((resolve, reject) => {

    let seatsAvailable = true;

    if(seatsAvailable)
        resolve("Enrollment Successful");
    else
        reject("Course Full");

});


enrollCourse
.then(msg => console.log(msg))
.catch(err => console.log(err));
```

Expected Output
Course: Web Technologies, Instructor: Dr. Kumar
Enrollment Successful

```
PS C:\Users\SYS\Desktop\Web_tecnologies_lab_work\Lab_08\Ex-3> node Course_enroll.js ...
Course: Web Technologies, Instructor: Dr. Kumar
Enrollment Successful
PS C:\Users\SYS\Desktop\Web_tecnologies_lab_work\Lab_08\Ex-3>
```

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